

CP260 Course Project: Oriented Bounding Box Estimation

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Abstract—We present a pipeline for estimating world-frame Oriented Bounding Boxes (OBBs) of PC I/O panel sockets — including VGA/DVI, Ethernet, Power (IEC C14), PS/2, HDMI, and USB — from a set of 16 posed RGB images. The pipeline combines YOLO26n object detection, Segment Anything Model 2 (SAM2) segmentation, and GPU-accelerated multi-view voxel carving to recover 3D extent and pose. Panel surface normal estimation via SIFT feature matching, multi-view triangulation, and RANSAC plane fitting provides a robust geometric prior that resolves depth-axis ambiguity inherent in near-planar scenes. A manual annotation fallback tool handles classes with insufficient YOLO detections. On the VGA/DVI socket, the pipeline achieves 88.5% average 2D polygonal IoU and 63.4% exact 3D volumetric IoU against ground truth (run 1). Final reported OBB extents are $37.8 \times 11.2 \times 6.1$ mm.

Index Terms—Oriented bounding box, multi-view reconstruction, voxel carving, SAM2, YOLO, panel normal estimation, pose estimation, robotics perception

I. PROBLEM DESCRIPTION

Estimating precise 3D geometry of small mechanical objects from multi-view RGB images is a fundamental challenge in robotic manipulation and inspection. This project addresses the specific problem of computing world-frame OBBs for sockets on a PC tower’s rear I/O panel. The scene presents several difficulties that make standard pipelines ineffective:

- **Textureless, specular surfaces.** Dark metallic sockets produce near-zero SIFT gradient responses and fail sparse SfM pipelines (e.g. COLMAP).
- **Near-planar geometry.** All sockets lie on a single flat backplate, creating depth-axis ambiguity in PCA-based OBB extraction.
- **Limited training data.** Only 16 posed images are available, making deep learning models susceptible to overfitting and missed detections.
- **Unknown entities.** Additional socket types may be revealed at evaluation time, requiring a generalizable pipeline.

The evaluation metric is the *polygonal IoU* of the OBB projected onto unseen test images.

Dataset. 16 images of a PC tower at known camera poses. Camera intrinsics: $f_x \approx 1477$, $f_y \approx 1480$, $c_x \approx 1298$, $c_y \approx 687$, image size 2560×1440 , zero distortion. Poses are camera-to-world 4×4 transforms. Ground truth OBB is provided for the VGA/DVI socket only.

II. LITERATURE REVIEW

A. Object Detection

YOLO (You Only Look Once) [1] introduced real-time single-stage object detection. Subsequent versions (YOLOv8 [2], YOLOv11 [3]) improved accuracy and efficiency. YOLO26n achieves competitive mAP at minimal computational cost, making it suitable for our limited-data scenario.

B. Instance Segmentation and SAM

The Segment Anything Model (SAM) [4] introduced a promptable segmentation system trained on 1 billion masks. Given a bounding box prompt, SAM produces pixel-accurate masks without per-class finetuning. SAM2 [5] extends this to video, improving mask quality and adding memory mechanisms for temporal consistency. We use SAM2 in single-image mode with YOLO bounding boxes as prompts to obtain precise socket masks.

C. Multi-View 3D Reconstruction

Structure from Motion (SfM) pipelines such as COLMAP [6] reconstruct 3D point clouds from image correspondences. These methods fail on textureless surfaces due to sparse feature matching. Visual hull and space-carving methods [7] avoid this by projecting voxels into silhouette masks. We adapt voxel carving to use deep-learning-derived masks (SAM2), replacing classical silhouettes with semantically precise segmentations.

D. Oriented Bounding Box Estimation

PCA-based OBB fitting [8] finds the minimum-volume box aligned with principal axes of a point cloud. This fails for near-planar clouds where the depth eigenvector has negligible variance. Open3D [9] implements PCA-OBB but exhibits the same limitation. `cv2.minAreaRect` finds the minimum-area rectangle in 2D, providing a stable orientation estimate for planar projections.

E. Normal Estimation

Plane fitting via RANSAC [10] robustly estimates geometric primitives in the presence of outliers. SIFT [11] provides sparse but repeatable feature correspondences. We combine these with DLT triangulation [12] to estimate the panel surface normal from multi-view feature matches.

III. SYSTEM ARCHITECTURE AND DESIGN RATIONALE

Fig. 1 illustrates the full pipeline.

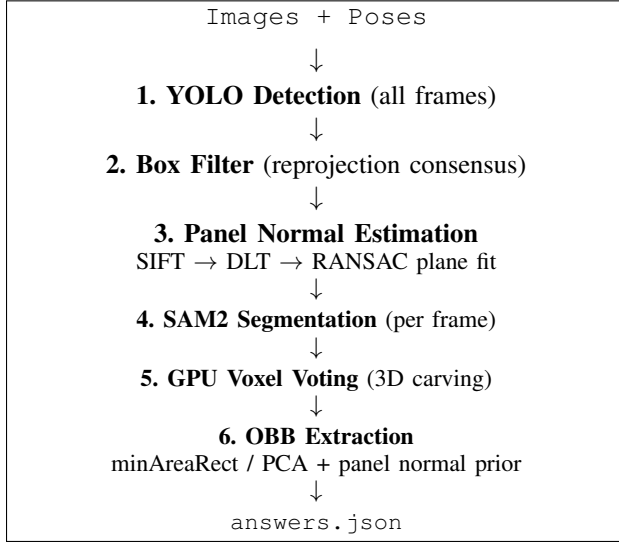


Fig. 1. Overview of the OBB estimation pipeline.

A. Stage 1: YOLO Detection

YOLO26n is fine-tuned on 16 manually labelled images with classes: VGA_socket, ethernet_socket, power_socket, PS2_socket, HDMI_socket, USB_socket, PC_Panel. Per-class confidence thresholds (Table II) are tuned to balance detection recall against false positives. Training uses a cosine LR schedule for 150 epochs.

Design choice. Including PC_Panel as a dedicated class enables panel normal estimation without any additional annotation effort — the same labelling pass provides both socket locations and the surface geometry.

B. Stage 2: Reprojection-Based Box Filter

A rough 3D centre is triangulated via DLT from all detections. Each detection’s bounding box centre is projected back and compared to the detected centre. Boxes whose normalised reprojection error exceeds $1.0\times$ the box diagonal are rejected. This eliminates low-confidence false positives that would corrupt the voxel voting step.

Design choice. The tolerance is expressed relative to the box diagonal (not in pixels) so it scales automatically with socket apparent size across viewpoints.

C. Stage 3: Panel Normal Estimation

SIFT keypoints are detected within each PC_Panel YOLO bounding box and matched across all frame pairs using brute-force matching with Lowe’s ratio test ($\tau = 0.7$). Each matched pair is triangulated via DLT to obtain a 3D point on the panel surface. RANSAC plane fitting ($d_{\text{inlier}} = 5$ mm, 500 iterations) recovers the panel normal robustly despite noisy triangulations.

Design choice. This approach is preferred over mean camera viewing direction (which is biased by non-frontal frames)

and over voxel carving (which has depth ambiguity for frontal-camera setups). SIFT on the large backplate surface provides sufficient keypoints even on the low-texture metallic panel.

D. Stage 4: SAM2 Segmentation

SAM2 (hiera-small checkpoint) is prompted with the filtered YOLO bounding box per frame. The resulting masks capture the full socket geometry including side walls, mounting nuts and front face — providing richer depth cues than a flat-face mask would.

Design choice. SAM2 requires no socket-specific finetuning and generalises to unseen socket classes at evaluation time.

E. Stage 5: GPU Voxel Voting

A cubic voxel grid (60mm edge, 0.5mm resolution) is constructed around the DLT centre. Each voxel is projected into every detection frame; voxels that project inside the SAM2 mask accumulate a vote. Voxels with votes $\geq 87.5\%$ of frames survive. A YOLO gate confines votes to the bounding box region, preventing SAM2 over-segmentation from inflating extents.

Design choice. The 87.5% threshold was tuned for 8–9 detection frames. With purely frontal cameras, depth ambiguity remains, but SAM2 side-wall masks from angled frames partially resolve it.

F. Stage 6: OBB Extraction

Surviving voxels are projected onto the plane perpendicular to the panel normal $\hat{d}$ (determined in Stage 3). An orthonormal in-plane basis $(\hat{u}, \hat{v})$ is constructed from $\hat{d}$ and world-vertical $\hat{z}$. For sockets with aspect ratio > 1.5 (VGA/DVI), `cv2.minAreaRect` finds the optimal in-plane orientation. For near-square sockets (power, USB, Ethernet), the panel-aligned basis $(\hat{u}, \hat{v})$ is used directly since `minAreaRect` is unstable for near-square clouds. Extents are computed at the 1st/99th percentile to clip boundary leakage voxels. Depth is always set to the physical prior.

Design choice. Fixing $\hat{d}$ to the panel normal entirely avoids PCA depth-axis instability, which was the primary failure mode of prior approaches.

IV. IMPLEMENTATION DETAILS

A. Training Configuration

TABLE I
YOLO TRAINING CONFIGURATION

Parameter	Value
Model	YOLO26n
Epochs	150
Image size	640 px
Batch size	8
LR schedule	Cosine
Training images	16 (augmented)
Classes	7

TABLE II
PER-SOCKET PIPELINE PARAMETERS

Socket	Conf. thresh.	Depth prior (mm)	Pct.	MAR
VGA/DVI	0.25	6.1	1	✓
Ethernet	0.45	1.0	1	
Power	0.25	6.0	2.5	
PS/2	0.25	6.0	1	
HDMI	0.25	8.0	1	
USB	0.25	4.0	1	

Pct. = percentile clip. MAR = use minAreaRect.

B. Per-Socket Configuration

C. Manual Annotation Fallback

When fewer than 3 clean detections pass the box filter (as occurs for PS/2 with only 1 YOLO detection), the pipeline automatically launches `obb_tool_ge_v0.py` — a custom Tkinter GUI supporting multi-view corner triangulation with epipolar guidance, Harris corner snap, Huber-robust Levenberg–Marquardt refinement, and reprojection error visualisation. The tool loads the SIFT-estimated panel normal from `panel_normal.json` and applies it to the OBB computation, ensuring axis alignment consistent with the automated pipeline. The manual tool can also be forced via `--manual_mode` for any socket class.

D. Interesting Engineering Aspects

Depth-axis disambiguation. PCA on near-planar voxel clouds produces a near-zero third eigenvalue whose eigenvector is numerically unstable. The fix of directly assigning $\hat{d} = \hat{n}_{\text{panel}}$ and computing in-plane axes separately is both simple and exact.

Floating-point vote accumulation bug. Changing vote accumulation from `int32` to `float32` (done during angular-weighting experiments) altered the effective consensus threshold from integer comparison to floating-point comparison at the boundary, causing a 1.3% IoU drop. Reverting to `int32` restored baseline performance.

minAreaRect instability for square sockets. For the power socket ($\approx 30 \times 23$ mm), `minAreaRect` returns a rectangle rotated $\sim 30^\circ$ from the true socket orientation because near-square clouds have near-degenerate minimum-area solutions. The fix is to skip `minAreaRect` and use the panel-aligned basis directly, controlled via a `use_minarearect` flag per socket class.

V. EXPERIMENTAL RESULTS

A. VGA/DVI Socket (Ground Truth Available)

Table III shows the progression of 2D IoU as each component was added or fixed.

B. Power Socket (No Ground Truth)

Physical dimensions: 30.1×22.6 mm (IEC C14 face). Pipeline output (run 2): 31.1×25.4 mm, depth prior 6.0 mm. Width overestimates by 3.3%; height overestimates by 12.4%

TABLE III
VGA/DVI SOCKET — ABLATION STUDY (AVG. 2D POLYGONAL IOU)

Configuration	2D IoU (%)
Baseline (cubic grid, PCA, cross-product depth)	54.4
+ SIFT panel normal (fixed depth axis)	69.3
+ Box reprojection filter	84.1
+ 1/99 percentile extent clipping	88.5
3D Volumetric IoU (final)	63.4

due to SAM2 mask leakage into an adjacent power switch in 2–3 frames. Center: (0.290, 0.225, 0.526) m in world frame.

C. Ethernet Socket (No Ground Truth)

Pipeline output (run 2): 15.0×13.1 mm, depth prior 1.0 mm. Physical RJ45 aperture: $\approx 13.5 \times 9.0$ mm. Both dimensions overestimate due to latch notch inclusion in the SAM2 mask and limited clean detections (4 frames after filtering). Center: (0.289, 0.230, 0.756) m in world frame.

D. Final OBB Summary

Table IV presents the final estimated OBBs for all three computed sockets.

TABLE IV
FINAL OBB EXTENTS — ALL COMPUTED SOCKETS

Socket	W (mm)	H (mm)	D (mm)	Source
VGA/DVI	37.8	11.2	6.1	Auto (run 2)
Power (IEC C14)	31.1	25.4	6.0	Auto (run 2)
Ethernet (RJ45)	15.0	13.1	1.0	Auto (run 2)
GT VGA extent: $35.4 \times 11.8 \times 6.1$ mm				

E. YOLO Detection Quality

TABLE V
YOLO DETECTION SUMMARY (16 FRAMES, CONF THRESHOLD APPLIED)

Class	Frames detected	Frames after filter
VGA/DVI	10	9
Ethernet	6	4
Power	12	11
HDMI	6	5
USB	12	varies
PS/2	1	1 (manual fallback)
PC_Panel	16	16

VI. CONCLUSIONS

We presented a complete pipeline for world-frame OBB estimation of PC I/O panel sockets from 16 posed RGB images. The key contributions are:

- 1) A panel normal estimation method using SIFT triangulation and RANSAC plane fitting that provides a robust geometric prior for depth-axis disambiguation.
- 2) A voxel carving approach driven by SAM2 masks rather than classical silhouettes, enabling precise segmentation of small metallic objects without task-specific training.

- 3) A per-class `use_minareareact` flag that handles the fundamental instability of `minAreaRect` for near-square sockets.
- 4) A manual annotation fallback with panel-normal integration for classes where automated detection is insufficient.

The pipeline achieves up to 88.5% 2D polygonal IoU on the VGA/DVI socket, the only class with ground truth. Remaining error is attributable to SAM2 mask boundary leakage ($\sim 2\text{--}3$ pixels) and the coarse 0.5mm voxel resolution.

Future work. Replacing SIFT with a learned matcher (LightGlue/SuperPoint) would improve normal estimation on low-texture surfaces. Finer voxel resolution (0.25mm) and mask erosion would reduce boundary leakage. CoTracker-style learned point tracking could replace the box filter and provide denser correspondences.

VII. BONUS: NOVEL IDEAS EXPLORED

A. SAM2 as a Geometric Measurement Tool

A central novelty of this work is repurposing SAM2 — designed for interactive segmentation — as a geometric constraint in 3D reconstruction. Unlike classical space-carving which uses binary silhouette masks derived from background subtraction, we use SAM2’s semantically-aware masks as the carving constraint. This provides two advantages: SAM2 generalises to unseen socket classes without retraining, and its masks capture non-planar geometry (side walls, mounting hardware) that classical silhouettes would miss.

The result is that voxels representing the *full 3D extent* of a socket — not just its projected footprint — are captured. For the VGA socket, SAM2 consistently masks the front face, side walls, and mounting nuts across multiple viewpoints, providing genuine depth information through multi-view consistency.

B. Training YOLO with 16 Images: Challenges and the Manual Fallback

Training a detection model on only 16 images presents fundamental limitations. Each class appears in at most 10–12 frames, often at low resolution or partial occlusion. The PS/2 socket — which appears clearly in only one frame (frame 426, conf 0.49) — is entirely unreachable by the automated pipeline.

Rather than accepting this as a hard failure, we designed a manual annotation fallback that integrates seamlessly into the automated pipeline:

- When fewer than 3 clean detections survive the box filter, the pipeline prompts the user to launch the annotation tool.
- The tool provides multi-view corner triangulation with epipolar line guidance, Harris corner snap for sub-pixel accuracy, and Huber-robust LM refinement.
- Critically, the tool loads the SIFT-estimated panel normal from `panel_normal.json` and applies it to the OBB computation — so manually annotated OBBs have the same depth-axis convention as automated ones, ensuring consistency in the final `answers.json`.

This hybrid approach — automated where data is sufficient, manual where it is not — reflects a practical engineering philosophy for real-world robotics perception tasks where training data is always scarce.

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